



Course:	Complex Variables and Transforms	Course Code:	MT 220
Program:	EE	Semester:	Fall-17
Duration:	60 min.	Total Marks:	25
Paper Date:	18-09-2017	Weight	12.5%
Section:	All sections	Page(s):	4
Exam:	Mid 1	Roll No:	

Instruction/Notes: Exchange of calculators or programmable calculators are not allowed at all. Attempt both questions on this question paper. Extra sheet may be used for rough work/calculations (Do not attach). Do all steps neat & clear.

1. What is harmonic function? Prove that the function $v = xy$ is harmonic. Also find the corresponding conjugate u and hence the original analytic function $f(z) = u + iv$, where $i = \sqrt{-1}$ (2+2+4+2=10 Marks, CLO-2)

(a) Harmonic functions.

A function $f(z)$ is said to be harmonic function in Domain " D " if the function is analytic in the Domain " D " and

- (i) Second derivatives exist in Domain
- (ii) satisfy the Laplace's equation

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(b) $v = xy$

Sol:-

$$V_x = y \Rightarrow V_{xx} = 0$$

$$V_y = x \Rightarrow V_{yy} = 0$$

$$\nabla^2 v = V_{xx} + V_{yy} = 0$$

Hence, The function is harmonic.

As we know that C-R Equations:-

$$u_y = -v_x \rightarrow (1)$$

$$u_x = v_y \rightarrow (2)$$

From Equation (2)

$$u_x = v_y = x$$

$$\Rightarrow u_x = x$$

$$\int u_x dx = \int x dx$$

2. The impedance of a circuit is given by the complex number Z satisfying the equation $e^Z = \frac{7i+1}{1+i}$.

- Solve the above equation to find the resistance (real part of impedance) and reactance across an inductor (imaginary part of impedance).
- Find the polar form of the given complex number.
- Using the answers of part (a) or part (b), calculate the ratio of the voltage difference amplitude to the current amplitude (magnitude of the complex number) and the phase difference between voltage and current (the argument of the complex number).
- Draw an argand diagram to give a pictorial description of the answers to the above part.

(7+3+3+2=15 Marks, CLO-1, 3)

(a) $e^Z = \frac{7i+1}{1+i}$

Sol:-

$$e^Z = \frac{7i+1}{1+i}$$

$$e^Z = \frac{7i+1}{1+i} \times \frac{1-i}{1-i}$$

$$e^Z = \frac{(7i+1)(1-i)}{1-i^2}$$

$$e^Z = \frac{8+6i}{2}$$

$$e^Z = 4+3i$$

where

$$Z = x+iy$$

$$e^Z = e^{x+iy}$$

$$e^{x+iy} = 4+3i$$

$$|e^Z| = e^x = 5$$

$$e^x = 5$$

$$x = \ln 5$$

$$e^x (\cos y + i \sin y) = 4+3i$$

$$e^x \cos y + i e^x \sin y = 4+3i$$

$$\tan y = \frac{\sin y}{\cos y} = \frac{3}{4}$$

$$y = \tan^{-1} \frac{3}{4}$$

$$Z = x+iy$$

$$Z = \ln 5 + i \tan^{-1} \frac{3}{4}$$

$$\begin{aligned} \operatorname{Re}(Z) &= \ln 5 \\ \operatorname{Im}(Z) &= \tan^{-1} \frac{3}{4} \end{aligned}$$